

Kafka Connect - Core Concepts

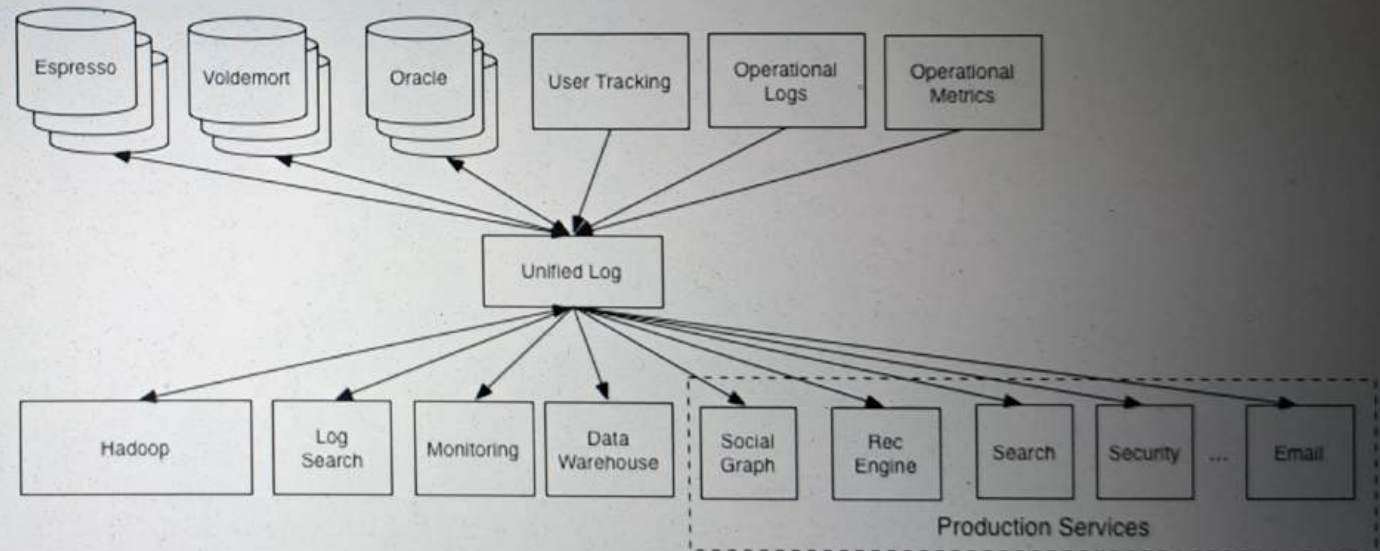


1. What is Kafka Connect?
2. How Kafka Connect works?
3. Kafka Connect Scalability
4. Kafka Connect Transformations
5. Kafka Connect Architecture

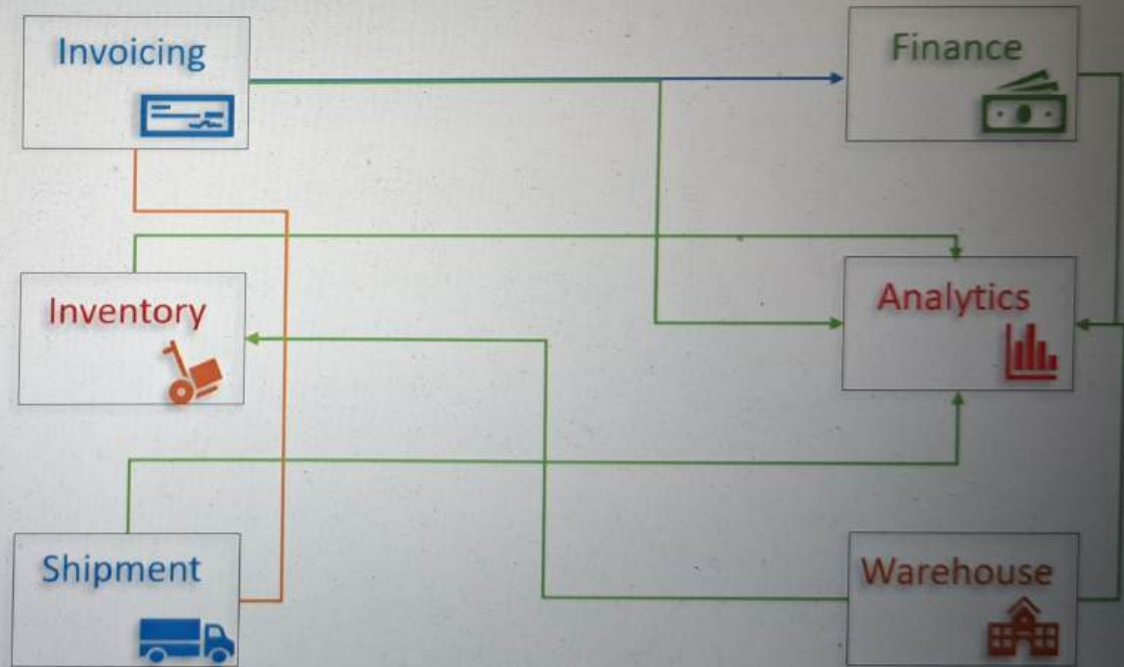
What is Kafka Connect?

Kafka Connect - Introduction

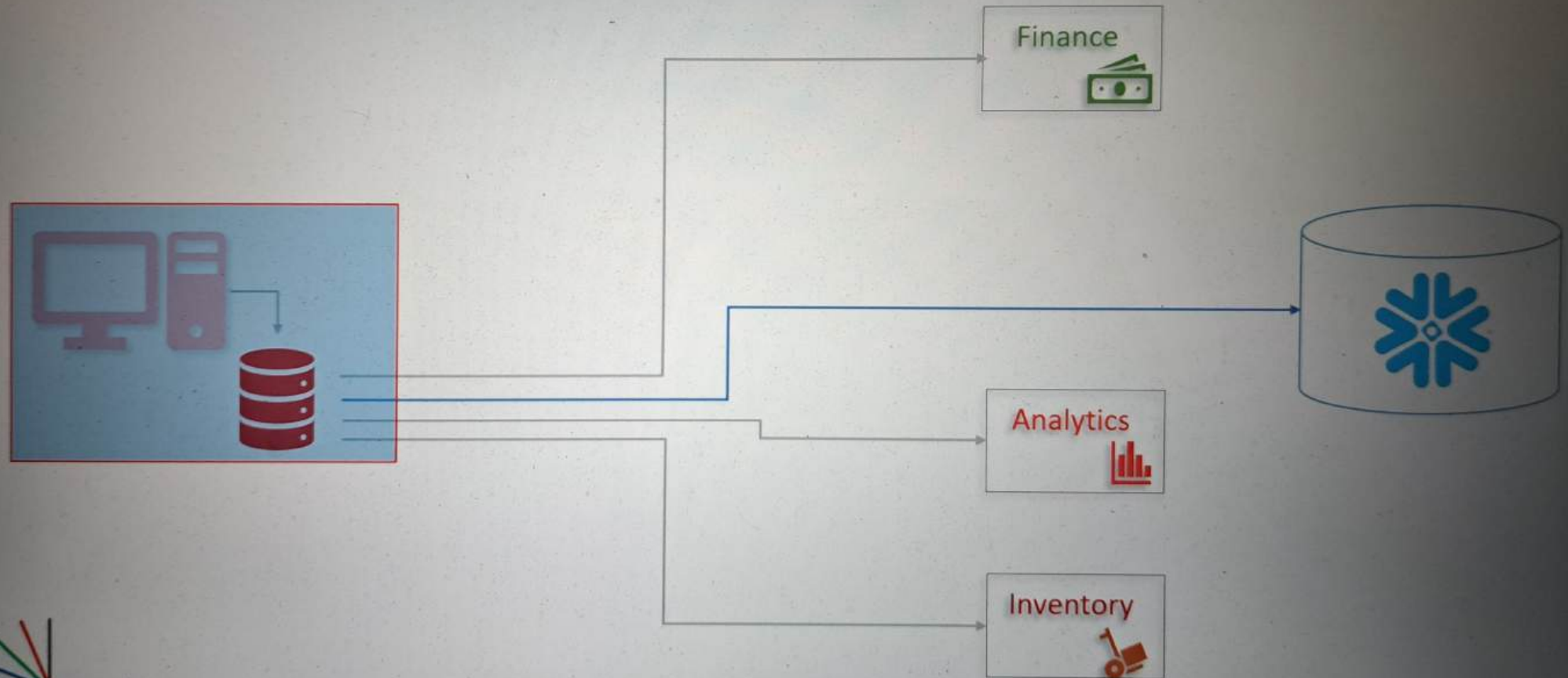
Created at  and open sourced in 2011



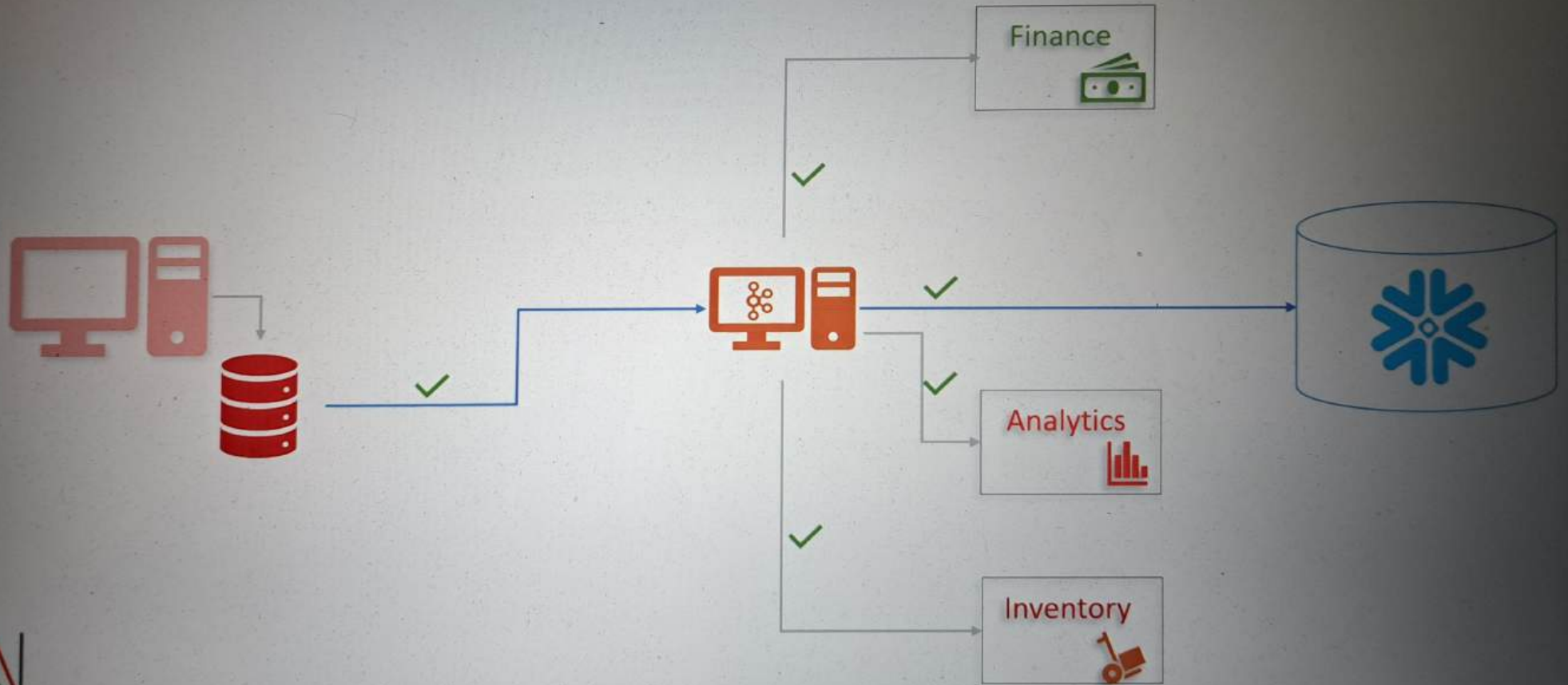
Kafka Connect - Introduction



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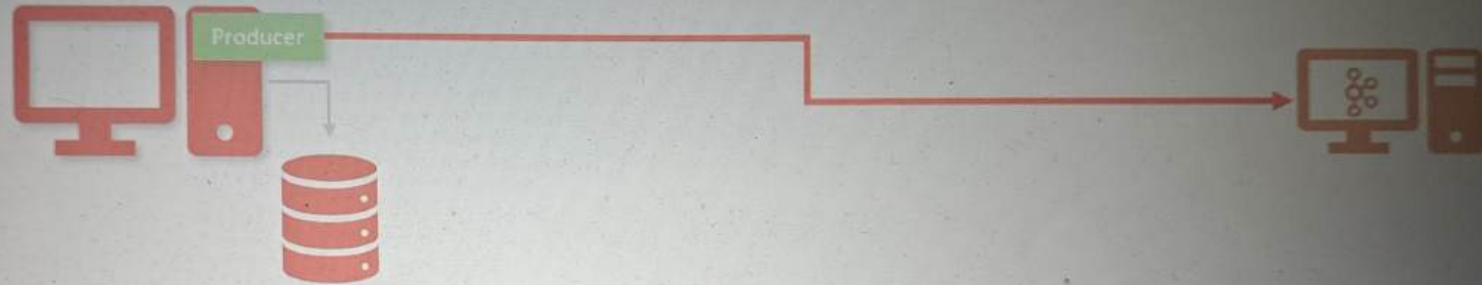


Kafka Connect - Introduction

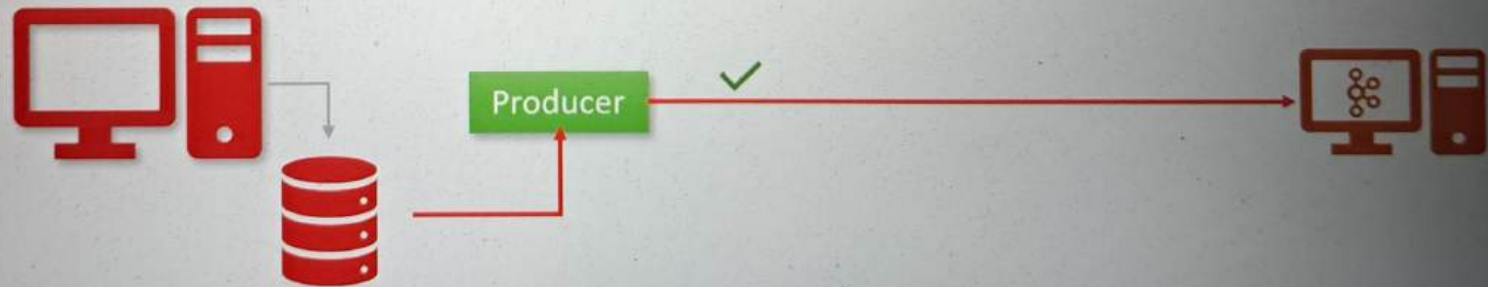


Kafka Connect - Introduction

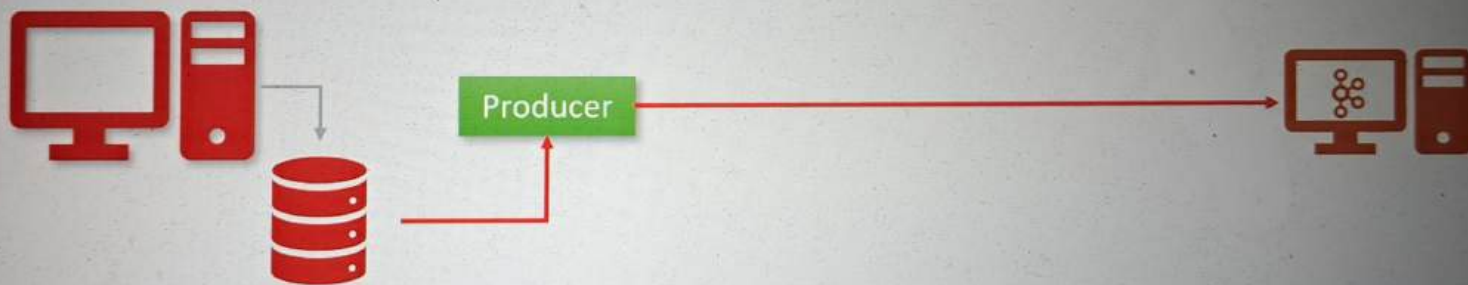
Option 1



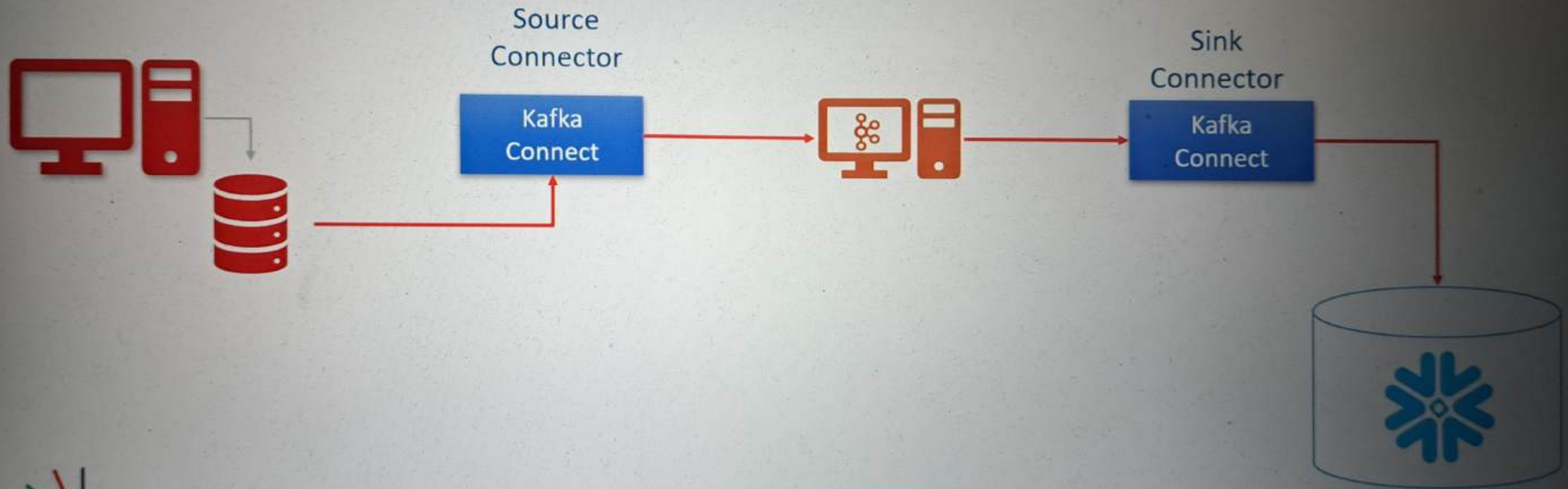
Option 2



Kafka Connect - Introduction

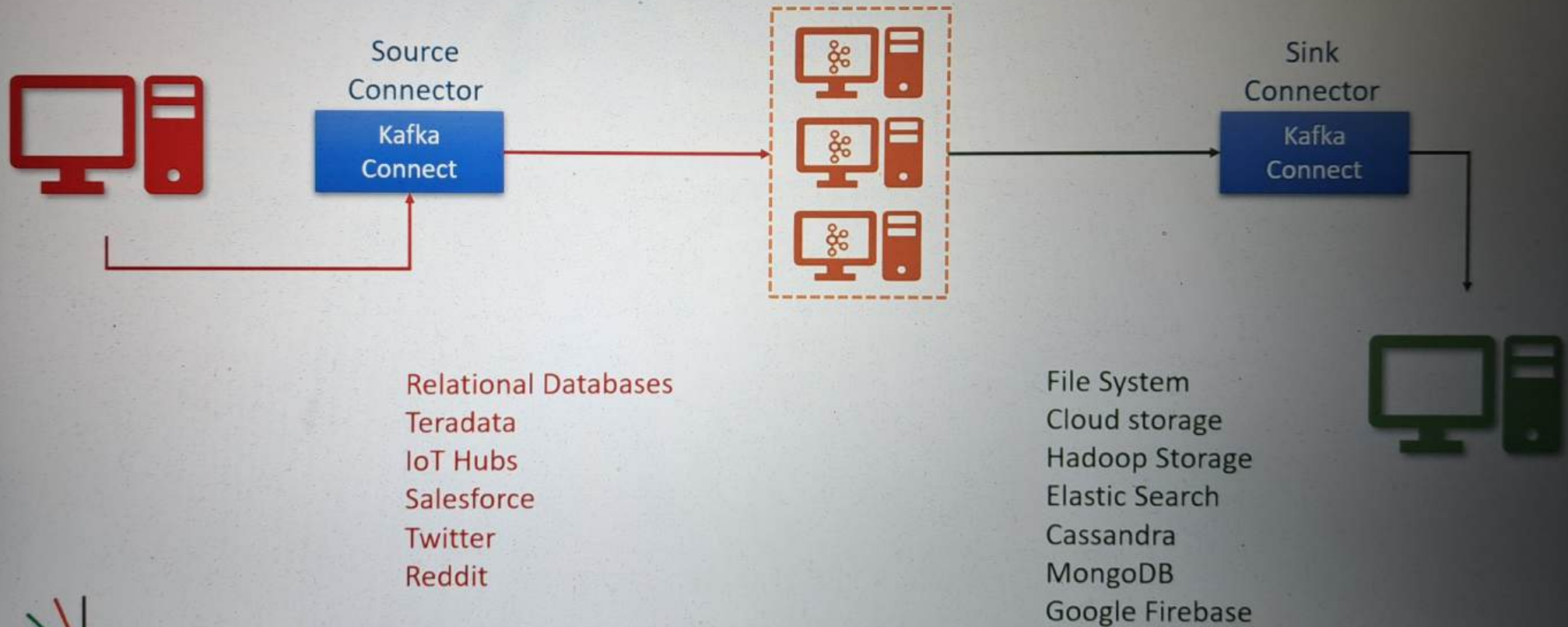


Kafka Connect - Introduction



How Kafka Connect works?

Kafka Connect – How it works?





Kafka Connect Framework

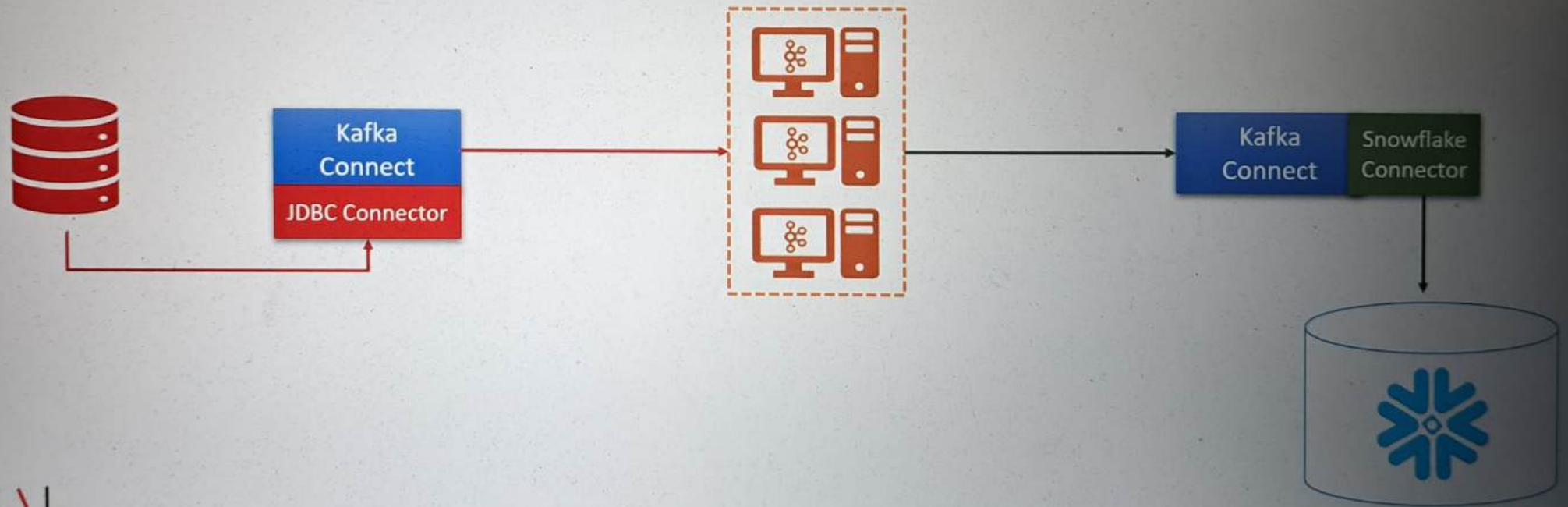
1. Source Connector

- i. SourceConnector
- ii. SourceTask

2. Sink Connector

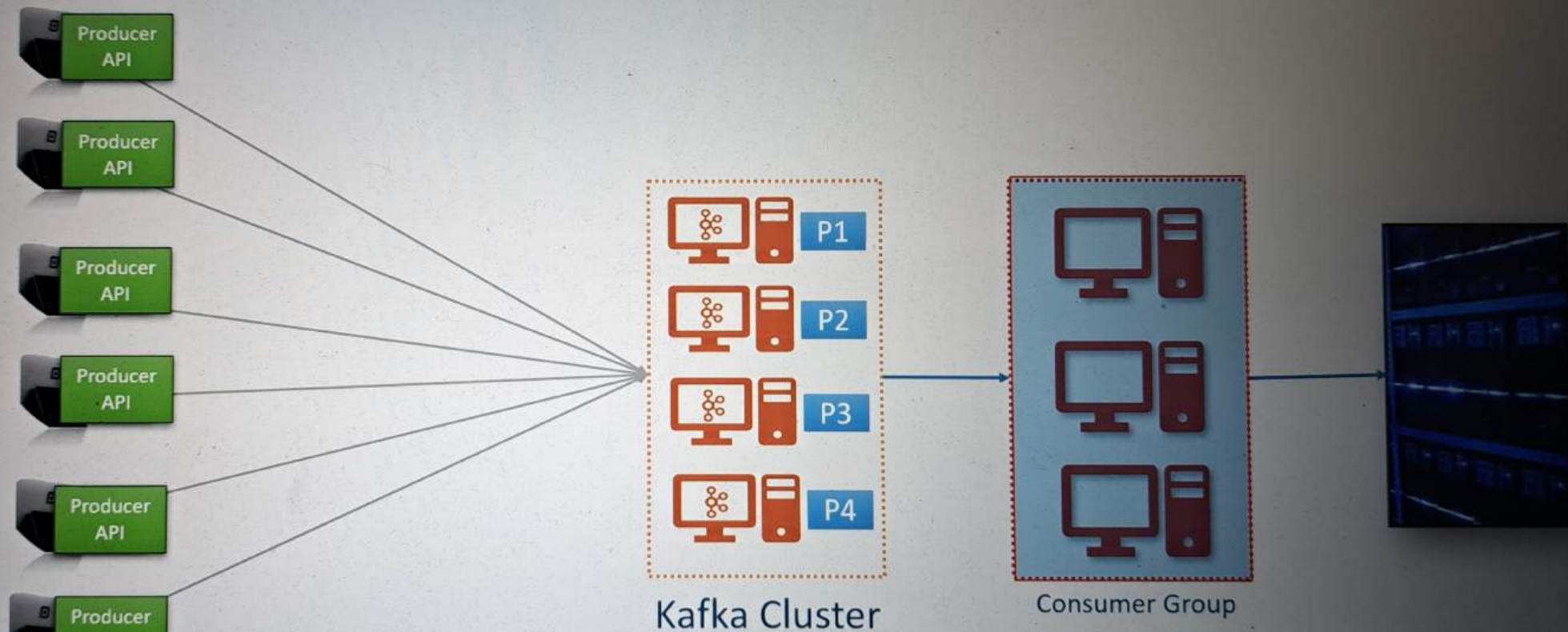
- i. SinkConnector
- ii. SinkTask

Kafka Connect – How it works?

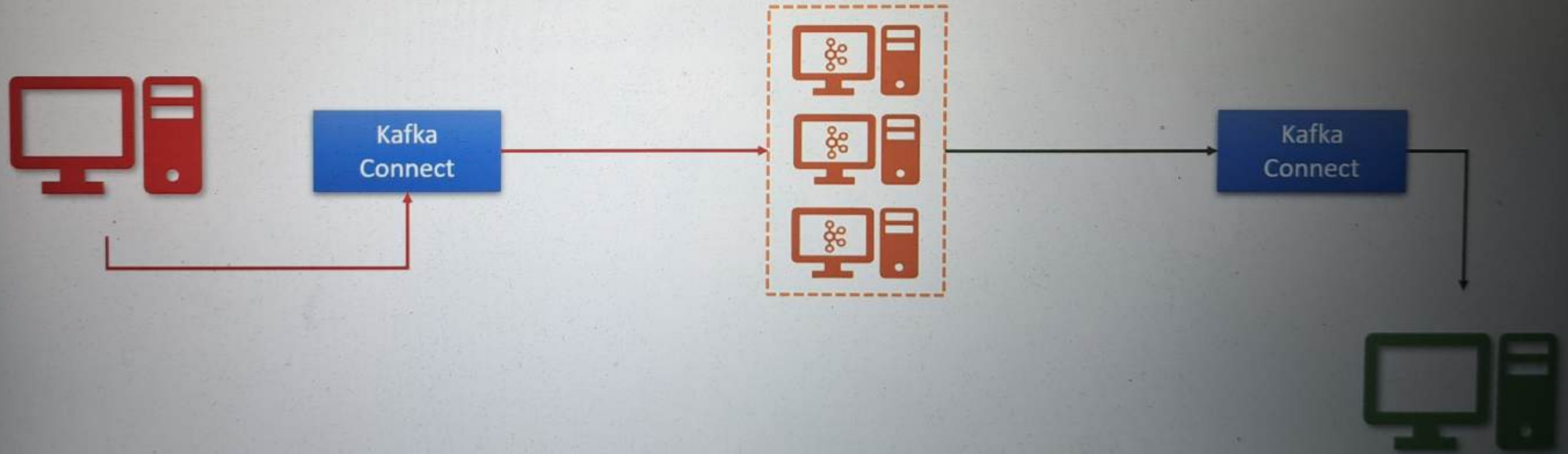


Kafka Connect Scalability?

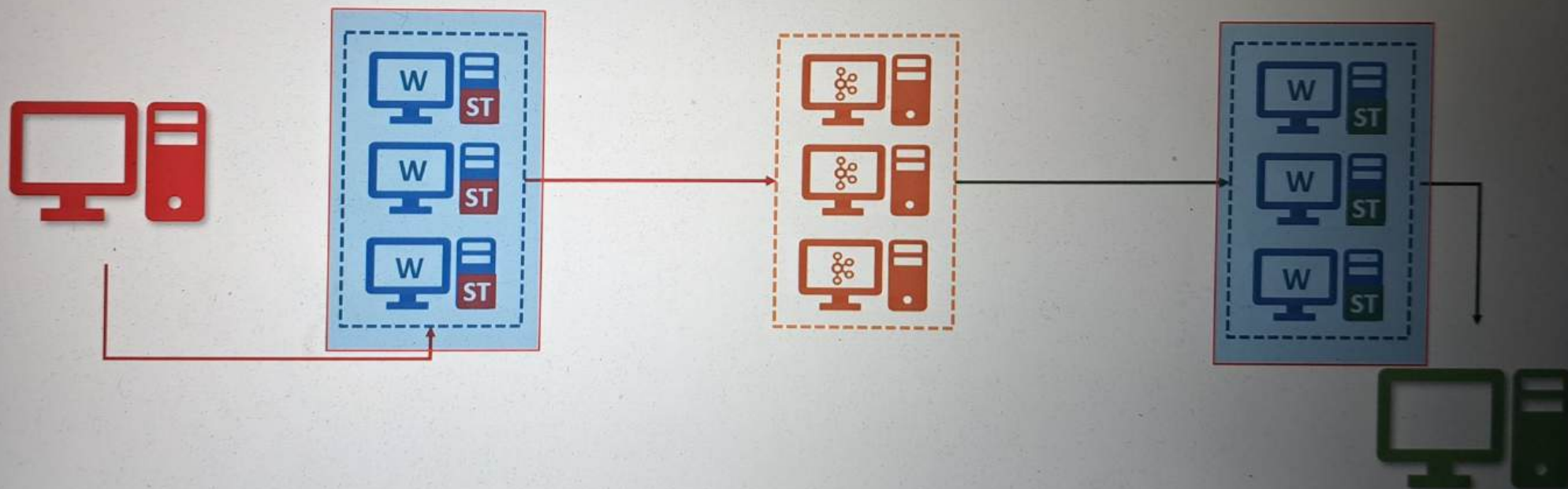
Apache Kafka – Scalability



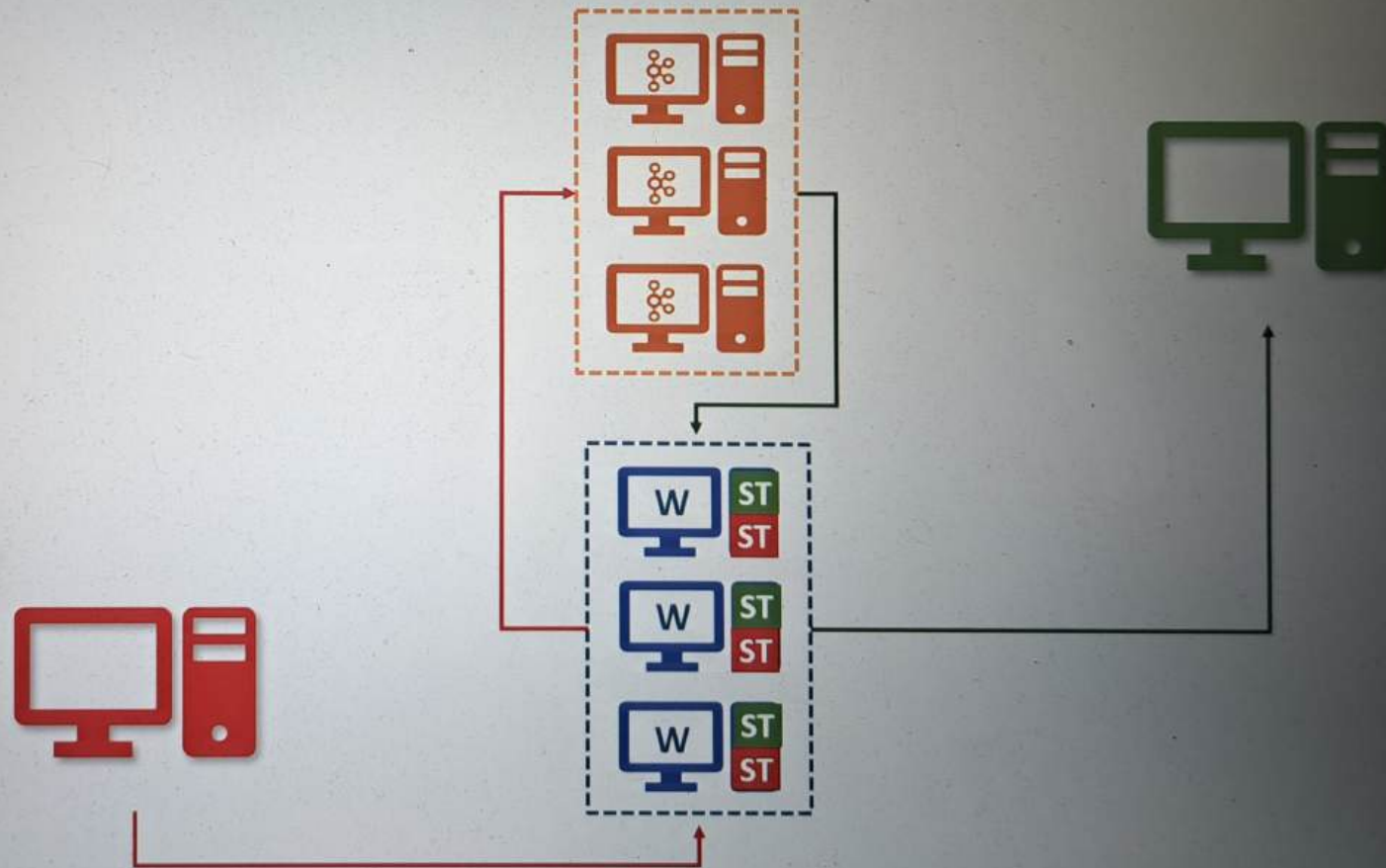
Kafka Connect – Scalability?



Kafka Connect – Scalability?



Kafka Connect – Scalability?

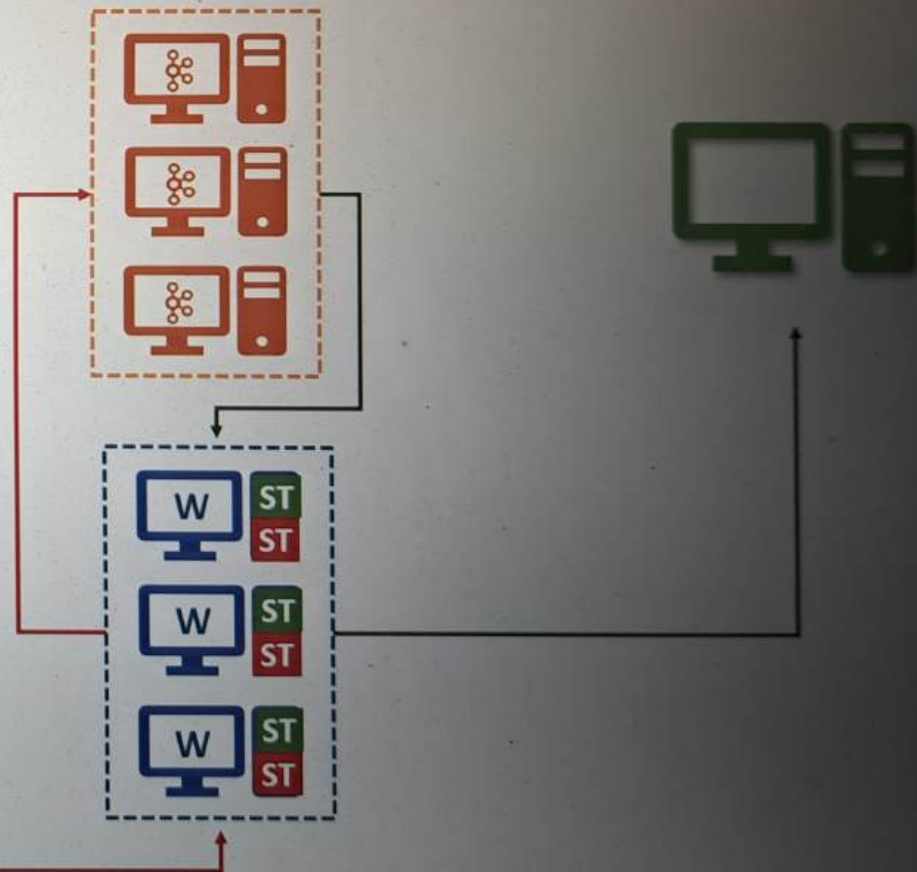


Connect Transformations?

Kafka Connect – Transformations?

Single Message Transformations – SMTs

1. Add a new field in your record using static data or metadata
2. Filter or Rename Fields
3. Mask some fields with a Null Value.
4. Change the Record Key
5. Route the record to a different Kafka Topic



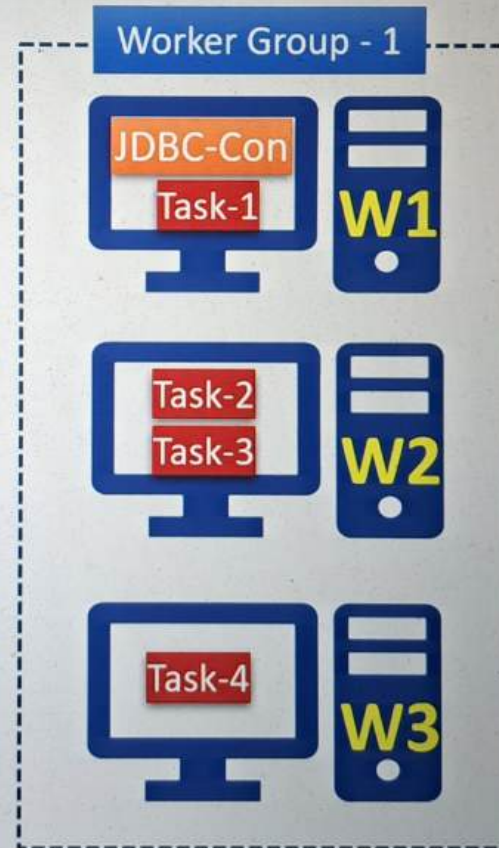
Kafka Connect Architecture?

Kafka Connect – Architecture?



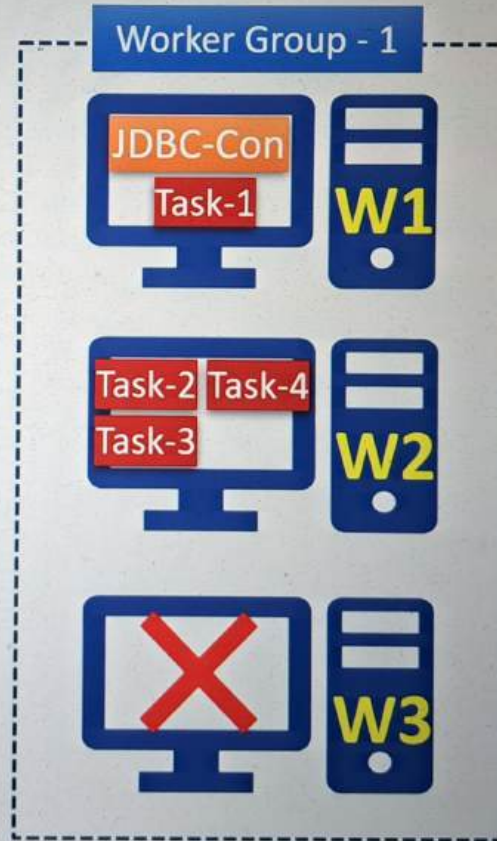
1. Worker
2. Connector
3. Task

Kafka Connect – Architecture?



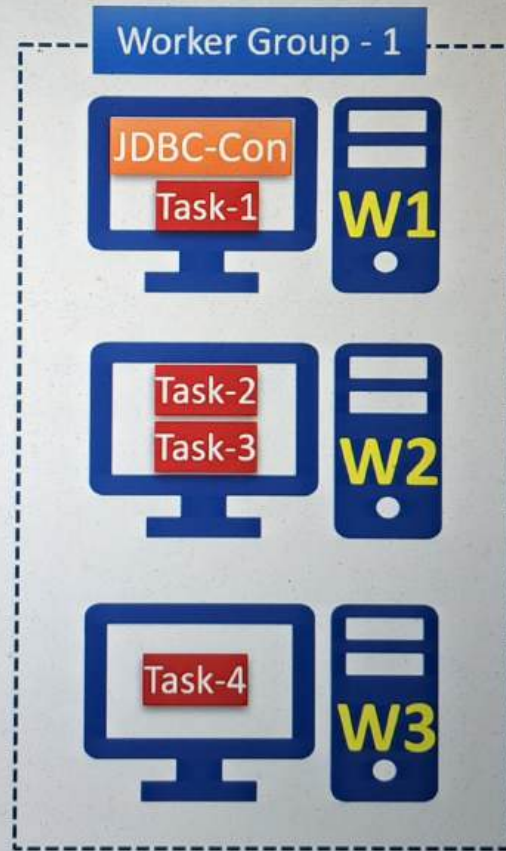
1. Fault Tolerant
2. Self Managed

Kafka Connect – Architecture?



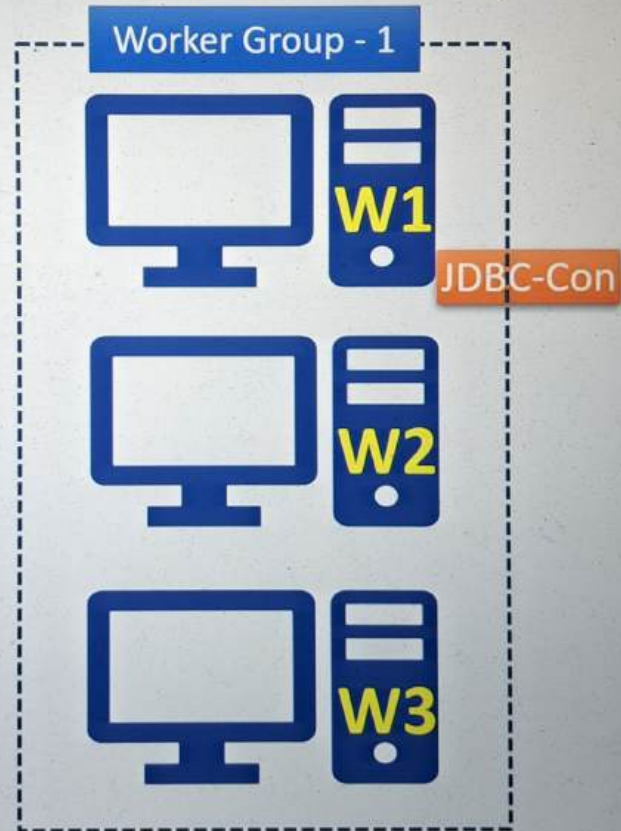
1. Fault Tolerant
2. Self Managed

Kafka Connect – Architecture?

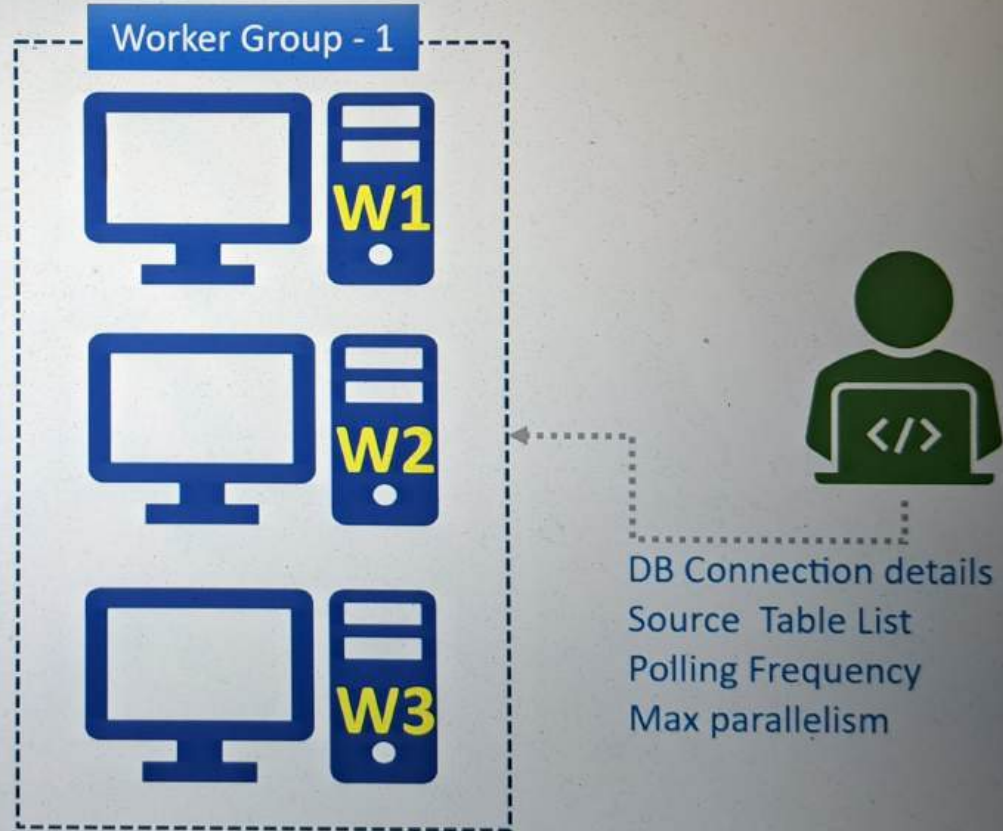


1. Reliability
2. High Availability
3. Scalability
4. Load Balancing

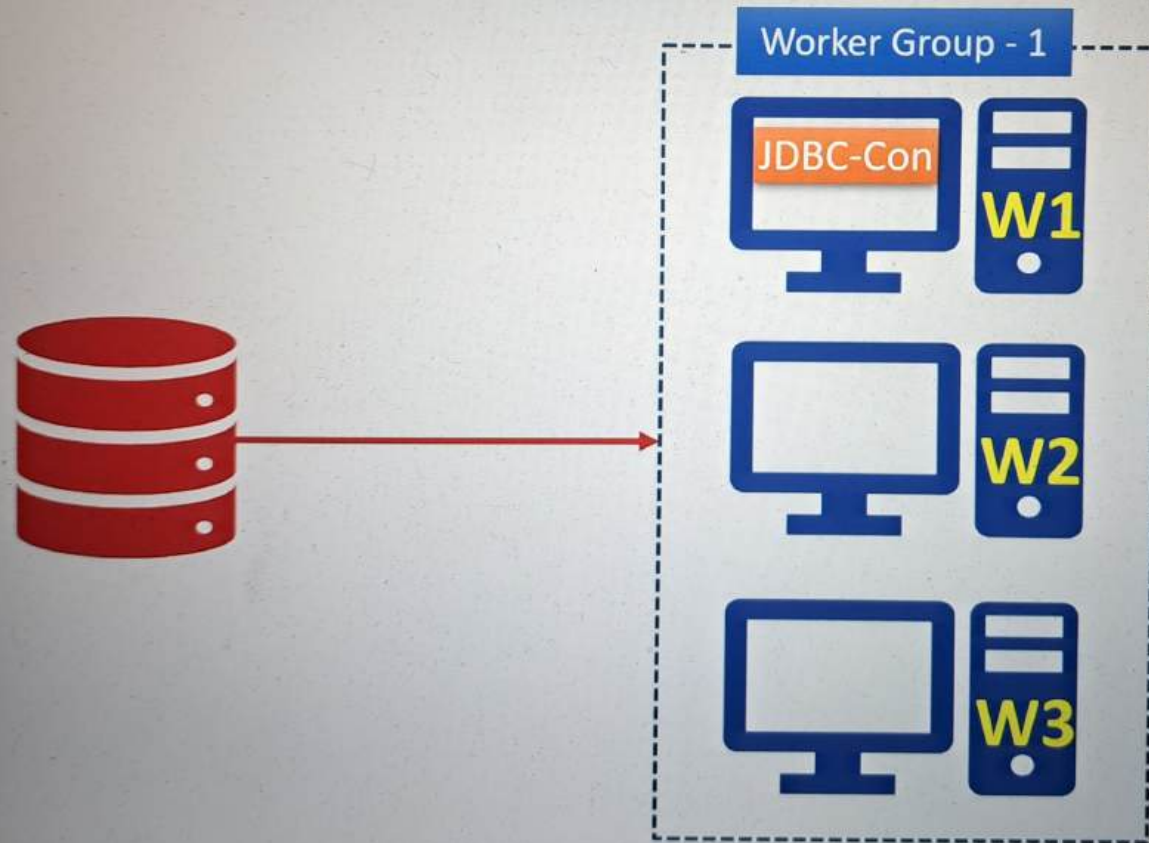
Kafka Connect – Architecture?



Kafka Connect – Architecture?

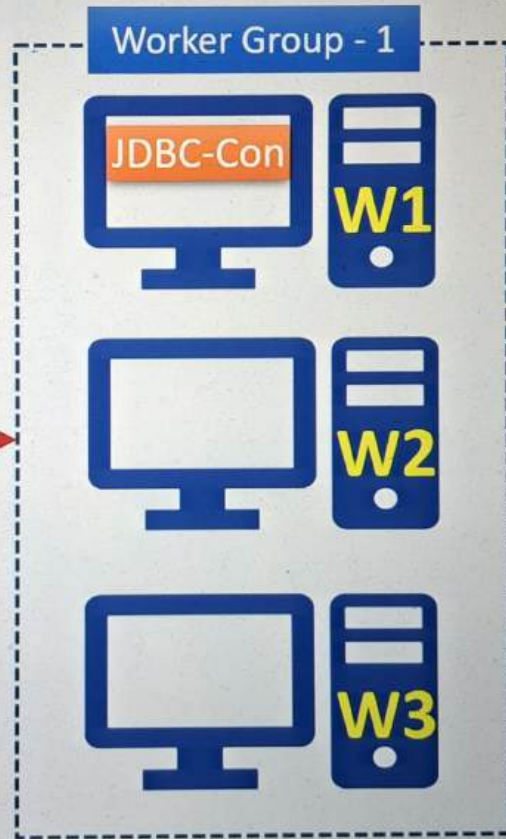


Kafka Connect – Architecture?



1. Task Spilt
2. Task List & Configuration

DB Connection details = XXXX
Source Table List = **T1, T2, T3, T4, T5**
Polling Frequency = 5 Min
Max parallelism = **5**

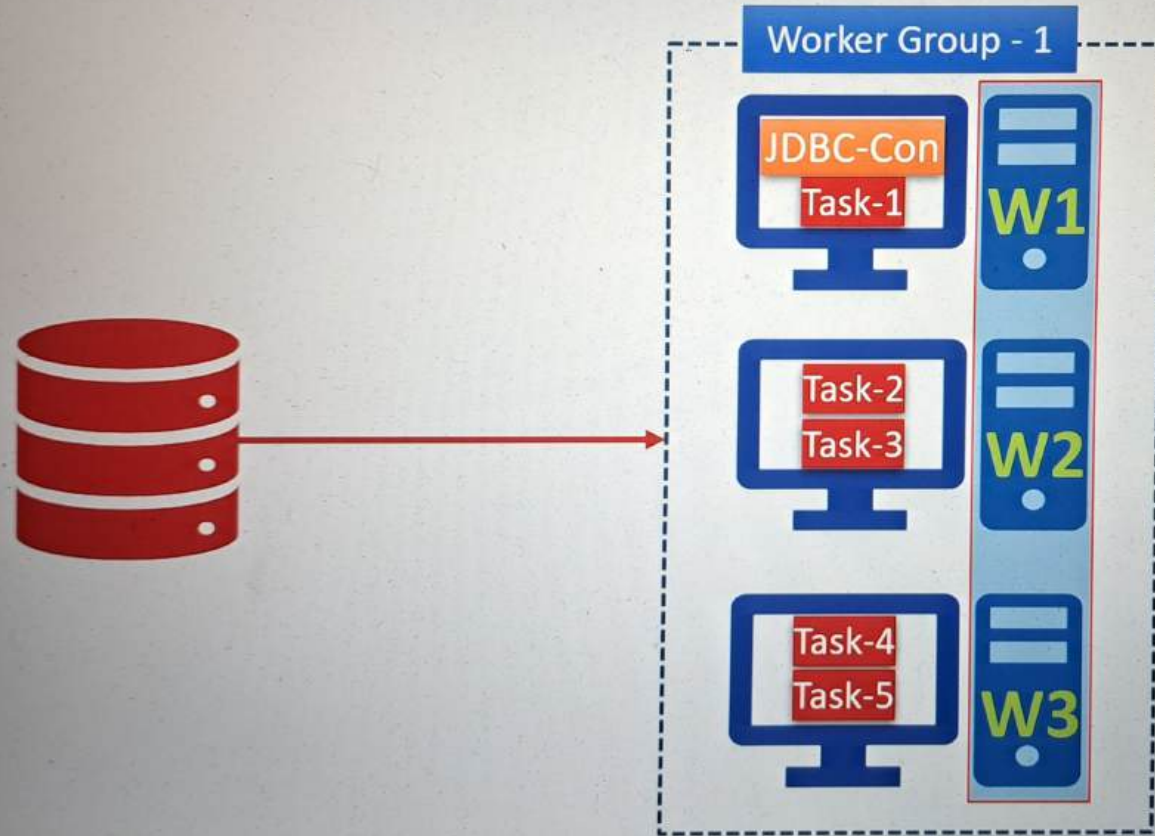


1. Task Spilt
2. Task List & Configuration

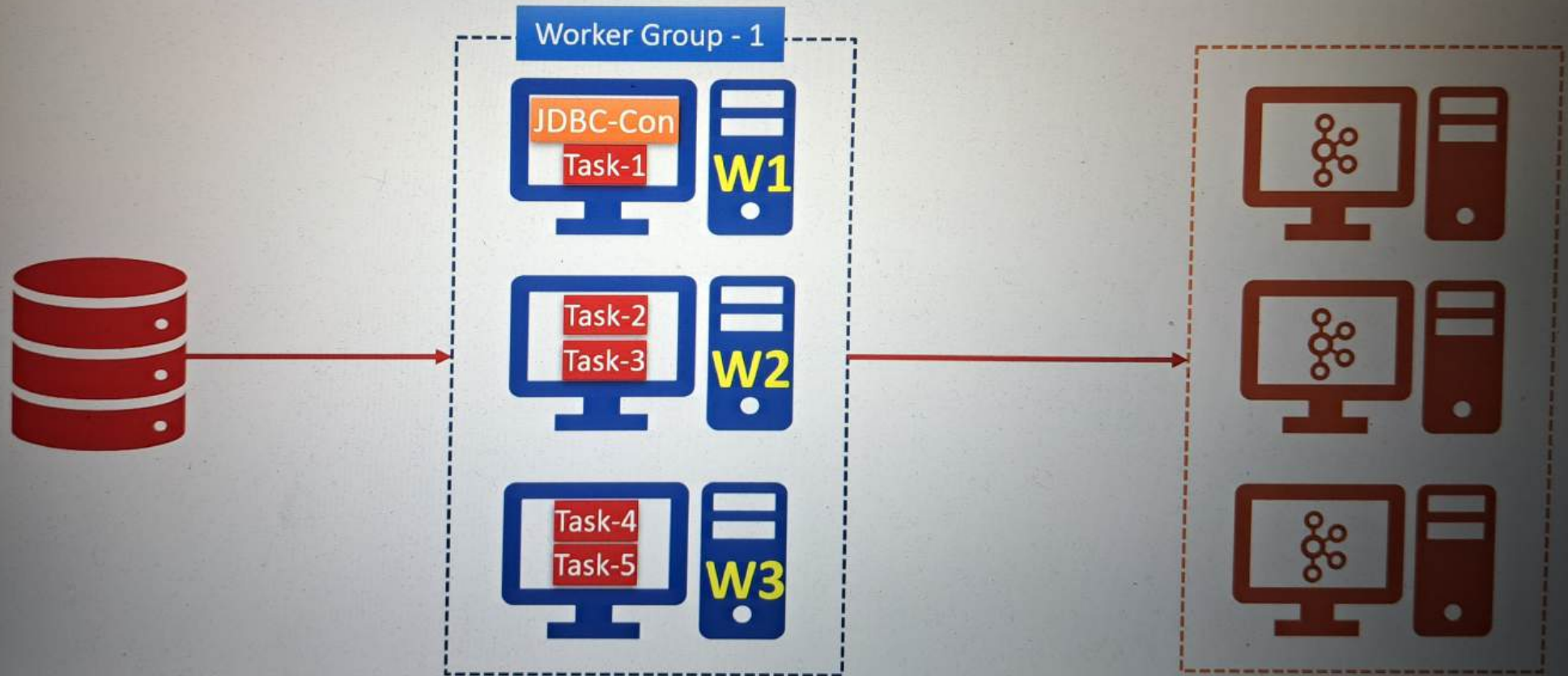
Task Name= Task1
Task Name= Task2
Task Name= Task3
Task Name= Task4
Task Name= Task5

Task Configs = XXX
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Kafka Connect – Architecture?



Kafka Connect – Architecture?



How to split the input?
How to interact with the external system?